Lingineni Venkata Karthik

Python Backend Developer |GenAIEngineer|FastAPI · LangChain·LangGraph · RAG · OpenCV · PyTorch · Azure

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Python · FastAPI · GenAI · LangChain · LangGraph · RAG Pipelines · LLMs · Vector Embeddings · NLP · OpenCV · InsightFace · PyTorch · Face Recognition · REST API · RBAC · JWT · MySQL · MS SQL Server · Azure Blob Storage · Linux · Docker · CI/CD · React.js · Django · PySpark

PROFESSIONAL SUMMARY

Python backend and **GenAI** engineer with 2 years of production experience building scalable **REST APIs**, enterprise systems, and AI-powered applications using **FastAPI**. Hands-on experience developing **RAG pipelines**, **LangChain** and **LangGraph**-based AI agents with intelligent tool routing, and integrating **LLMs** into backend workflows. Delivered a production **Face Recognition System** using **OpenCV**, **InsightFace**, and **PyTorch**. Builtautomated trading platforms, GST accounting systems, and student admission portals serving **500+ users**. Experienced with **Azure Blob Storage**, **RBAC**, SQL optimisation, and Linux server deployments. Available to join**immediately**.

EXPERIENCE

Python Backend Developer — GenAI & Enterprise Systems | Lingsaa Technologies PvtLtd

Guntur,India | May2024–Aug2026

- Designedanddeployedscalable RESTful APIs using Python (**FastAPI**) for 3 production enterprise applications—serving500+ concurrent users with sub-200ms response time, **JWT** authentication,and role-basedaccesscontrol(**RBAC**).
- DevelopedanAIchatbotagent using **LangChain** and **LangGraph** — implements intent-based toolrouting via LangGraphnodes,using **RAG pipelines** (**vector embeddings** + retrieval) for read queries and dedicatedCRUDtoolsfor data modification.
- Built andintegratedaproduction Face Recognition System using **OpenCV**, **InsightFace**, **PyTorch**,and **SilentFaceAnti-Spoofing** — real-time liveness detection to prevent photo/video spoofing attacks.
- Applied**NLP**fundamentals — **transformers**, tokenization, and embedding models — to design andtune **RAG**retrievalpipelines,improving **LLM** response accuracy and reducing hallucination in production.
  - Reduceddatabasequerylatency by ~40% through indexing strategies, normalised schema designacross **MySQL**and**MSSQLServer** — diagnosed using slow query logs, EXPLAIN analysis, and **SAR reports**.
- Integrated**AzureBlobStorage** for scalable media and document management, replacing local filestorage and enablingreliablecross-server file access in production.
- IntegratedNTTDataPayment Gateway, broker APIs, SendGrid, SMSCountry, and WhatsApp Business API — automatingtransaction workflows and reducing manual overhead significantly.
- Managed**Linux**production deployments using Git, **Nginx**, and PM2 — maintained 99%+ uptime; diagnosed performanceissuesusing **SAR**, htop, and Nginx error logs.
- Built **React.js**frontendsfor all 3 enterprise systems — dashboards, data entry forms, and reporting views — enablingfullend-to-enddelivery.

PROJECTS

AI Chatbot Agent with LangGraph Tool Routing

2025 – Present

- Architected a **FastAPI** backend where **LangGraph** orchestrates user intent — routing retrieval queriesto a **RAG pipeline** and data modification requests to dedicated CRUD tool nodes.
- Built **vector embedding** pipeline for document ingestion and similarity-based retrieval — reducing unnecessary **LLM** inference calls and improving response latency.
- Applied **transformer** architecture, tokenization strategies, and embedding distance metrics to optimise **RAG pipeline** retrieval accuracy and minimise hallucination.

⚙️**Python · FastAPI · LangChain · LangGraph · RAG · VectorEmbeddings · NLP · LLMs| In development**

Face Recognition System with Anti-Spoofing (FRS)

2025

- Built real-time FRS pipeline: **OpenCV** for frame capture, **InsightFace** for identity matching, Silent Face Anti-Spoofing for liveness detection, and **PyTorch** for deep learning model inference.
- Replaced manual identity verification in production — reducing verification time to under 2 seconds per check and eliminating human error in identity checks.

⚙️**Python · OpenCV · InsightFace · SilentFaceAnti-Spoofing · PyTorch| Integrated in production**

LingsaaAITraders —AutomatedTrading Platform *2024*

- Engineered real-time tick data ingestion pipeline aggregating 1-minute OHLC candles —processing thousands of market events per minute with zero data loss using optimised async **FastAPI** workers.
- Built algorithmic trading strategy engine with automated broker API execution — reducingtrade execution from minutes to milliseconds.
- Optimised financial analytics pipelines using **Pandas** and **NumPy** — cutting query time by35%.

⚙️**Python · FastAPI · Node.js · MSSQLServer · Pandas · NumPy | Codeavailableonrequest**

ANUCDEDistance EducationPortal *2024*

- Built complete backend for student admission platform handling 1,000+ enrolments — REST APIs, payment gateway integration, automated PDF generation, reducing manual processing by 60%.
- Developed **React.js** admin dashboard with real-time tracking and exportable reports — eliminating manual spreadsheet reporting.

⚙️**Python · FastAPI · React.js · MySQL · Pandas | Codeavailableonrequest**

Finance & Accounts Management System with GST Filing *2025 – 2026*

- Architected multi-role accounting backend with **RBAC** across 5 user roles — enforcing strict data segregation per role.
- Built GST filing API with OTP verification — cutting average filing time from hours to under10 minutes per submission.

⚙️**Python · FastAPI · Node.js · MSSQLServer · Pandas · NumPy | Codeavailableonrequest**

ExpenseTracker —FastAPI+ React (Open Source Demo) *2025*

- Full-stack demo with JWT auth, RBAC, real-time dashboard — containerised with **Docker** and deployed via **GitHub Actions CI/CD** pipeline.

⚙️**Python · FastAPI · React.js · MySQL · Docker · GitHubActions | github.com/venkatakarthik705/expense-tracker**

TECHNICAL SKILLS

Python Backend: Python 3.12, FastAPI, REST API Design, JWT Authentication, RBAC, SQLAlchemy, Node.js

Frontend: React.js, HTML5, CSS3, Bootstrap, JavaScript

GenAI & LLMs: LangChain, LangGraph, RAG Pipelines, Vector Embeddings, NLP, Transformers, Tokenization, LLMs

Computer Vision: OpenCV, InsightFace, Silent Face Anti-Spoofing, PyTorch, Face Recognition (FRS)

Databases: MySQL, MS SQL Server, SQL Optimisation, Indexing, Schema Design, Pandas, NumPy

Cloud & Storage: Azure Blob Storage

DevOps & Tools: Git, Linux, Nginx, PM2, SAR, Docker, GitHub Actions (CI/CD), WinSCP

Integrations: NTT Data Payment Gateway, Broker APIs, SendGrid, SMSCountry, WhatsApp Business API

Familiar / Learning: TypeScript, Django, PySpark, Kubernetes, Microsoft Azure (AZ-900 — in progress)

EDUCATION

B.Tech in Computer Science Engineering — RVIT (Affiliated to JNTUK) *2020 – 2024*

Relevantcoursework:DBMS,OOP,ComputerNetworks, OperatingSystems

SSC — 92% — Viveka Public School *2018*